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KEYWORDS:

keyword1, keyword2, keyword3, keyword4

1 | INTRODUCTION

From confirmation report We now consider settings in which the extremal dependence structure is non-stationary, and may change over time. For example, meteorological variables observed across spatial locations may exhibit changes in extremal dependence due to shifts in climate regimes, while biomarker measurements collected longitudinally for individuals may display structural changes associated with disease progression, treatment effects, or lifestyle changes (????). In such settings, it is of interest to detect changepoints or structural breaks in the extremal dependence structure.

In the existing framework, the multivariate “groups” could, in principle, represent different time periods. However, this method was not developed to accommodate temporally evolving dependence structures within an already multivariate setting (e.g., spatial), and we assumed that the data within each group was stationary. This is limiting in applications where tail dependence is expected to vary over time. In environmental contexts, for example, climate change may strengthen or weaken extremal dependence across an entire spatial domain, as suggested by ?, who found that recent Southern European heatwaves have been more widespread than in the past. More generally, several types of structural change may occur. In some situations the extremal dependence structure may shift globally across all sites, corresponding to a regime change affecting the entire system. In other settings, only a subset of sites may change their extremal dependence behaviour, for example by switching between clusters of locations with similar tail dependence characteristics. Discarding the stationarity assumption and allowing for such temporal changes in extremal dependence is therefore a natural and important extension of our current work. It would enable us to identify time periods with distinct spatial patterns of tail dependence, as well as the specific changepoints at which these transitions occur. Therefore, a key focus of this work will be the development of methods for changepoint detection in multivariate extremal dependence structures, building on the divergence-based framework introduced in our first paper.

†This is an example for title footnote.
⁰**Abbreviations:** ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor

There has been some previous work on changepoint detection in multivariate dependence structures, whether in the context of extremes or more generally. For example, ? developed a non-parametric sliding window approach for detecting changepoints in pairwise extremal dependence. Also, ? developed a Bayesian clustering algorithm for multivariate extremes which utilises a mixture of Dirichlet distributions, and they show how it can be used to temporally cluster negative returns for several S&P 500 stock sectors, thereby identifying periods of time with similar extremal dependence structures across the sectors. In ?, a likelihood-based method for detecting temporal changepoints in spatio-temporal data is developed, and applied to daily wind speed data in Ireland, similar to the data we used in our first paper. We hope to build on these existing approaches and also the divergence-based clustering framework developed in our paper to develop a method for detecting changepoints in multivariate extremal dependence structures, which can be applied to a wide range of datasets and applications.

2 | METHODOLOGY

In this section, we describe a framework for detecting changepoints in multivariate extremal dependence structures. Section ?? introduces the CE framework of [Heffernan and Tawn \(2004\)](#) and ?, which we use to model the extremal dependence structure of multivariate data. Section ?? describes how the extremal skew-geometric JS^{G} divergence can be used to quantify the difference between two extremal dependence structures. Finally, in Section ??, we describe a permutation-based procedure for detecting changepoints in the extremal dependence structure over time, using the divergence measure to quantify differences between temporal blocks.

2.1 | Conditional extremes model for multivariate extremal dependence

We model multivariate extremal dependence using the conditional extremes (CE) framework of [Heffernan and Tawn \(2004\)](#) and ?. This framework describes the behaviour of a random vector conditional on one of its components exceeding a high threshold, and can accommodate both asymptotic dependence and asymptotic independence. We briefly summarise the spatial formulation used here; further details are provided in ?.

Following [Heffernan and Tawn \(2004\)](#), all vectorised operations are applied componentwise. Let $\mathbf{X}_s = (X_{1,s}, \dots, X_{d,s})$ denote a d -dimensional random vector observed at spatial location $s \in \{1, \dots, D\}$. Following ?, we transform each component to standard Laplace margins, yielding $\mathbf{Y}_s = (Y_{1,s}, \dots, Y_{d,s})$. This standardisation separates marginal behaviour from the extremal dependence structure and permits both positive and negative associations.

For conditioning variable $i \in \{1, \dots, d\}$ and location $s \in \{1, \dots, D\}$, the CE model is

$$(\mathbf{Y}_{-i,s} | Y_{i,s} = y) = \boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \mathbf{Z}_{|i,s}, \quad y > u_i, \quad (1)$$

where $\mathbf{Y}_{-i,s}$ denotes $\mathbf{Y}_s$ with its i^{th} component removed, u_i is a high threshold, and all vector operations are componentwise. The parameter vectors $\boldsymbol{\alpha}_{|i,s} \in [-1, 1]^{d-1}$ and $\boldsymbol{\beta}_{|i,s} \in (-\infty, 1]^{d-1}$ describe, respectively, the strength of extremal association and the conditional scaling of the remaining components. The residual vector $\mathbf{Z}_{|i,s}$ is assumed to be independent of $Y_{i,s}$ above u_i .

Following [Heffernan and Tawn \(2004\)](#), we adopt the working assumption

$$\mathbf{Z}_{|i,s} \sim \text{MVN}_{d-1}(\boldsymbol{\mu}_{|i,s}, \boldsymbol{\Sigma}_{|i,s}),$$

which gives

$$(\mathbf{Y}_{-i,s} | Y_{i,s} = y) \sim \text{MVN}_{d-1}(\boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \boldsymbol{\mu}_{|i,s}, \boldsymbol{\Omega}_{|i,s}(y)), \quad y > u_i, \quad (2)$$

where

$$\boldsymbol{\Omega}_{|i,s}(y) = \text{diag}(y^{\boldsymbol{\beta}_{|i,s}}) \boldsymbol{\Sigma}_{|i,s} \text{diag}(y^{\boldsymbol{\beta}_{|i,s}}).$$

The model is fitted separately for each conditioning variable, spatial location and temporal block, without imposing spatial smoothness. A common threshold u_i , defined on the standard Laplace scale, is used across locations to ensure that the resulting conditional distributions are comparable. The Gaussian working model also permits the closed-form divergence calculations described in Section ??.

2.2 | Extremal skew-geometric Jensen-Shannon divergence for multivariate extremal dependence

2.3 | Changepoint detection in non-stationary multivariate extremal dependence structures

Let $M(T)$ denote the estimated extremal dependence structure for a time block T , which could represent a given period of time for which we have sufficient data to fit the CE model and estimate the corresponding divergence matrix. A simple way to compare two time blocks T_1 and T_2 is via the statistic

$$D_{\text{obs}} = \|M(T_1) - M(T_2)\|_{\text{F}}, \quad (3)$$

i.e. the Frobenius norm of their difference, defined as

$$\|A\|_{\text{F}} = \sqrt{\sum_{i,j} A_{ij}^2}.$$

This statistic captures the overall or global magnitude of change in extremal dependence across all pairs of sites. Another option is the Infinity norm, defined as

$$\|A\|_{\infty} = \max_{i,j} |A_{ij}|,$$

which may better highlight the maximum difference between each matrix, placing greater emphasis on more localised changes in tail dependence. Critical values for this statistic could be generated via simulation from fitted models to assess whether observed changes represent significant shifts in tail dependence.

One key question is how large the observed statistic must be to indicate a significant change in extremal dependence. A natural way to assess whether the difference between two time periods reflects a genuine structural change, rather than sampling variability in the fitted CE models, is via a permutation test. A permutation test is a non-parametric method for testing hypotheses by comparing the observed test statistic to a reference distribution generated by randomly permuting the data (?). Such procedures are exact when the observations are exchangeable under the null hypothesis. One advantage of this approach is that it avoids strong parametric assumptions about the sampling distribution of the divergence matrix, which is unknown, and directly respects the dependence structure of the observed data.

2.3.1 | Permutation test for temporal changes in extremal dependence

Suppose we wish to test for a change in extremal dependence at a candidate split point τ by comparing two local time blocks immediately before and after the split. Let T_1 and T_2 denote the corresponding pre- and post-change windows, each of size n . We fit the CE model separately in each block, yielding divergence matrices $M(T_1)$ and $M(T_2)$ as defined in Equation (??) (or their aggregate form in Equation (??)).

Our null hypothesis is

$$H_0 : \text{the extremal dependence structure is identical in } T_1 \text{ and } T_2.$$

Under this null hypothesis of stationarity, exceedances from the two blocks arise from the same distribution. We therefore generate a reference distribution for D_{obs} by repeatedly permuting the time indices:

1. **Pool observations:** For a chosen block size $n < \min\{|T_1|, |T_2|\}$, combine the final n observations from time block T_1 and the first n observations from T_2 into a single dataset.
2. **Permute time labels:** Randomly permute the time indices of the pooled observations and reassign them to two pseudo-blocks of size n .
3. **Refit CE models:** For the permuted dataset, reapply the exceedance threshold u_i for each conditioning variable and refit the CE model in each pseudo-block, yielding divergence matrices $M^{(b)}(T_1)$ and $M^{(b)}(T_2)$, for $b = 1, \dots, B$ permutations.
4. **Recompute statistic:** Compute the test statistic

$$D^{(b)} = \|M^{(b)}(T_1) - M^{(b)}(T_2)\|_{\text{F}}, \quad b = 1, \dots, B. \quad (4)$$

Because the CE model fitting and divergence calculation are repeated within each permutation, this procedure naturally propagates the estimation uncertainty in the fitted dependence parameters, somewhat analogous to a bootstrapping approach.

The empirical p -value is then

$$\hat{p} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{D^{(b)} \geq D_{\text{obs}}\}. \quad (5)$$

A small value of $\hat{p}$ provides evidence that the tail-dependence structure differs between the two time periods. In performing the permutation test, a suitable block size n must be chosen to ensure reliable estimation of the CE model parameters within each time block, while also providing sufficient power to detect changes in extremal dependence. Using the same n across all permutations ensures that the sampling variability in the fitted parameters is appropriately reflected in the reference distribution of the test statistic.

2.3.2 | Sliding-window permutation approach for changepoint localisation

The permutation test above assesses whether a *given* time split corresponds to a change in extremal dependence. We now extend this idea to estimate the location of an unknown changepoint.

Given N total observations and a block size $n < N$, for each candidate changepoint $\tau \in \{n, \dots, N - n\}$, define the corresponding time blocks

$$T_1(\tau) = \{\tau - n + 1, \dots, \tau\}, \quad T_2(\tau) = \{\tau + 1, \dots, \tau + n\},$$

For each τ , we perform the permutation test described above to compute a p -value $\hat{p}(\tau)$ as in equation (3), for the null hypothesis of no change in extremal dependence at τ .

To avoid running computationally intensive permutation tests at every possible “split” point τ , we first identify promising candidate changepoint locations. To “screen” for these time points, we evaluate $D(\tau)$ from equation (2) over a grid of admissible split points from time points n to $N - n$. The coarseness of this grid can be adjusted to balance computational cost with the resolution of candidate changepoint detection. Local maxima of the function $D(\tau)$ indicate times at which the extremal dependence structure differs most strongly between the two sides of the split. Let $\{\hat{\tau}_1, \dots, \hat{\tau}_K\}$ denote the K largest local maxima, which we treat as candidate changepoint regions. The function $D(\tau)$ therefore acts as a screening statistic highlighting candidate changepoint locations.

Now that we have identified candidate changepoint locations $\{\hat{\tau}_1, \dots, \hat{\tau}_K\}$, we refine our inference by applying the permutation test described above within a local window around each candidate point.

$$\mathcal{W}_k = \{\tau : \hat{\tau}_k - w \leq \tau \leq \hat{\tau}_k + w\},$$

for some window half-width $w > 0$. For each $\tau \in \mathcal{W}_k$, we:

1. Define blocks $T_1(\tau) = \{\tau - n + 1, \dots, \tau\}$ and $T_2(\tau) = \{\tau + 1, \dots, \tau + n\}$, as above.
2. Compute the observed statistic $D_{\text{obs}}(\tau) = \|M(T_1(\tau)) - M(T_2(\tau))\|_F$.
3. Generate its permutation distribution under $H_0(\tau)$ (no change at τ).
4. Obtain a p -value $\hat{p}(\tau)$ as in equation (3).

This produces a local significance profile within each window $\mathcal{W}_k$, mapping τ to $\hat{p}(\tau)$.

Within each window $\mathcal{W}_k$, we define the most likely changepoint as

$$\tilde{\tau}_k = \arg \min_{\tau \in \mathcal{W}_k} \hat{p}(\tau),$$

i.e. the split point showing the strongest evidence against temporal homogeneity of extremal dependence. Equivalently, $\tilde{\tau}_k$ corresponds to the most statistically significant peak in $D(\tau)$ after accounting for sampling variability. Furthermore, if $\hat{p}(\tilde{\tau}_k) < \alpha$ for some significance level α , we reject the null hypothesis of no change at $\tilde{\tau}_k$, and conclude that there is evidence of a changepoint in extremal dependence at this time point.

This two-stage procedure separates effect size from statistical significance: the screening step identifies times at which the estimated extremal dependence structure changes most strongly, while the permutation step determines whether these changes are larger than would be expected under temporal homogeneity. To fully characterise the peak in $D(\tau)$ around $\tau = 1000$, this procedure can be repeated over a local window.

The proposed framework provides a flexible starting point for detecting temporal changes in multivariate extremal dependence structures. Several methodological questions remain open. In particular, further work is required to investigate the statistical power of the proposed discrepancy measures under different types of dependence change, to assess the impact of temporal dependence on the validity of permutation procedures, and to develop computationally efficient implementations suitable for large spatial networks. The power of this procedure will depend on both the magnitude of the change in extremal dependence and the number of observations available in each time block. Furthermore, in the presence of temporal dependence, block

permutations or other dependence-preserving resampling schemes may be required to ensure valid inference. Nevertheless, the approach outlined here offers a natural extension of the divergence-based clustering framework developed in the previous chapter and has the potential to provide new tools for identifying structural changes in extremal dependence across time.

3 | SIMULATION STUDY

We demonstrate the performance of our proposed changepoint detection procedure through a simulation study. To assess the ability of our method to detect changes in extremal dependence, we generate synthetic datasets with known changepoints designed to mimic the application to Spanish temperature and precipitation data in Section 4. **More emphasis on “gradual” change?**

4 | APPLICATION TO SPANISH DATA

- We now apply our changepoint detection algorithm to data from Spain.
- In Section 4.1, we introduce the dataset of temperatures and precipitation indexes used in our application.
- Section 4.3 shows the results of our initial screening step, while in Section 4.4, we discuss the results of our changepoint detection algorithm.
- Finally, Section 4.5 shows the clustering results for seasons before and after identified changepoints.

4.1 | Spanish Data

- We analyse temperature and precipitation observations from 40 meteorological stations distributed across mainland Spain.
- Their locations are shown in the left panel of Figure 1 .
- The analysis covers the period 1960-2020, inclusive.
- Only stations satisfying the required record-length and data-availability criteria are retained. **State criteria for choosing stations/locations (description in code).**
- Daily maximum temperature and precipitation totals are obtained from the European Climate Assessment Dataset (ECAD) ([European Climate Assessment & Dataset \(ECA&D\) n.d.](#)).
- The ECAD provides quality-controlled daily meteorological observations from stations across Europe.
- We aggregate the daily observations to a weekly resolution, taking the maximum recorded temperature and total accumulated precipitation within each week.
- This aggregation reduces the influence of very short-range temporal variation and decreases the proportion of zero precipitation observations, although it does not necessarily remove temporal dependence completely.
- The relationship between high temperature and dry conditions may depend more strongly on precipitation accumulated over preceding weeks than on precipitation observed during the same week.
- We therefore represent precipitation conditions using the Standardised Precipitation Index (SPI) of [McKee, Doesken, and Kleist \(1993\)](#), rather than using weekly precipitation totals directly.
- At each station, we calculate SPI from precipitation accumulated over a rolling 13-week window.
- Thus, the value assigned to a given week summarises precipitation during that week and the preceding 12 weeks, providing a measure of precipitation conditions over an approximately three-month period.
- The accumulated precipitation is modelled using a Gamma distribution, whose parameters are estimated using unbiased probability-weighted moments.

- Its fitted cumulative probability is then transformed to a standard Gaussian scale.
- Under the usual SPI convention, negative values represent drier-than-normal conditions and positive values represent wetter-than-normal conditions.
- For the dependence analysis, we therefore define the drought index [$D = -\text{SPI}$] so that larger values correspond to drier conditions.
- This sign convention means that high temperatures and dry conditions are expected to exhibit positive extremal dependence.
- The right-hand panels of Figure 1 illustrate the relationship between weekly maximum temperature and the drought index at Daroca, in Aragon, and Jerez de la Frontera, in Andalusia.
- These stations are highlighted in the map in the left-hand panel.
- At both stations, the distribution of weekly maximum temperature differs substantially between seasons.
- The seasonal separation in drought-index values is less pronounced at Daroca, whereas Jerez de la Frontera exhibits clearer seasonal variation in both temperature and drought conditions. **Mention explicit figures for means/95th quantiles??**
- These seasonal differences motivate fitting the dependence model and applying the changepoint procedure separately within each season.
- This allows both the marginal behaviour and the extremal dependence structure to vary seasonally.

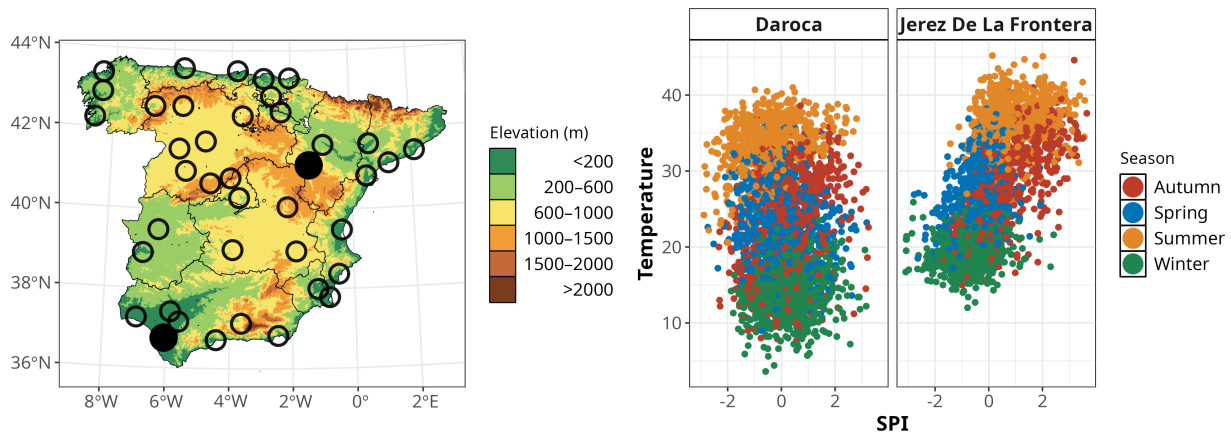


FIGURE 1 Left: Elevation (m) profile of mainland Spain along with locations of 40 meteorological stations. Daroca and Jerez de la Frontera highlighted. Right: Weekly maximum temperature against 13-week drought index, ($D = -\text{SPI}$), at these two stations, with observations coloured by season.

4.2 | Setup

- The conditional extremes framework is applied after transforming each variable to a common marginal distribution.
- Consequently, systematic changes in the marginal distributions over time could otherwise be confounded with changes in the extremal dependence structure.
- Both variables exhibit seasonal variation, while weekly maximum temperature in particular also displays a long-term marginal trend.

- We therefore preprocess the marginal distributions before fitting the conditional extremes models.
- At each station, and for each variable, we consider observations from each calendar month separately and fit a linear trend in year.
- The estimated trend is removed from the observations for that month.
- Treating months separately allows the magnitude of the temporal trend to vary throughout the year while accounting for the pronounced marginal seasonality.
- This detrending removes an estimated linear change in the location of each monthly marginal distribution; it does not imply that all temporal dependence has been removed.
- In particular, some short-range dependence may remain between successive weekly observations, and consecutive values of the 13-week drought index are constructed from overlapping precipitation windows.
- Following detrending, the observations are grouped by meteorological season.
- For each station and variable, the resulting seasonal observations are transformed standard Laplace margins using the empirical rank transform.
- The conditional extremes and changepoint analyses are then performed separately for winter, spring, summer and autumn.
- We fit the conditional extremes model of [Heffernan and Tawn \(2004\)](#) using a common dependence threshold equal to the theoretical 80th percentile of the standard Laplace distribution.
- The same threshold is used for both variables, at every station, in both temporal blocks and at every candidate changepoint.
- This ensures that exceedances have a common interpretation throughout the analysis.
- The 80th-percentile threshold provided a satisfactory empirical compromise between restricting attention to the joint tail and retaining enough exceedances for stable model estimation.
- In particular, the resulting clustering solutions exhibited clear spatial contiguity. **Add a sensitivity analysis for the dependence threshold.**
- The Monte Carlo approximation used to construct the pairwise dissimilarities employs the same sample from the standard Laplace distribution in every temporal comparison.
- Holding this sample fixed avoids introducing additional Monte Carlo variation between candidate changepoints.
- The conditional fits from the two conditioning directions are combined into the aggregated dissimilarity matrix defined in Section ?? **Test for others**
- All screening, testing and clustering calculations reported below use this matrix. **Confirm terminology and add sensitivity results for the alternative dissimilarity matrices, if required.**
- For the Spanish application, the permutation procedure is adapted to preserve the temporal and spatial structure within each season.
- Rather than permuting individual weekly observations, we permute complete season-years.
- For spring, summer and autumn, the season-year is the calendar year in which the observations occur.
- Winter spans two calendar years; December is therefore assigned to the same season-year as the following January and February.
- At a candidate season-year (t), the two temporal blocks consist of the (25) season-years ending at (t) and the (25) season-years beginning at ($t + 1$), respectively.
- Thus, (t) denotes a possible change between season-years (t) and ($t + 1$).

- In each pointwise permutation, the 50 season-years from the two blocks are pooled and randomly divided into two pseudo-blocks containing 25 season-years each.
- Every weekly observation belonging to a given season-year is moved as a single unit, and the same reassignment is applied simultaneously to both variables and all 40 stations.
- This preserves the dependence between observations within a season-year and the spatial correspondence between stations.
- The validity of this permutation procedure relies on treating season-years as exchangeable under the null hypothesis of no change in the extremal dependence structure.
- Although permuting complete season-years preserves within-year dependence, it does not preserve any residual dependence between successive season-years.
- The choice of block length involves a bias-variance trade-off.
- Shorter blocks provide better temporal localisation but contain fewer threshold exceedances, resulting in unstable or failed conditional extremes fits.
- Longer blocks provide more stable estimates but leave fewer feasible candidate changepoints and smooth changes over a wider temporal interval.
- We investigated block lengths of 15, 20, 25 and 30 season-years.
- With 15-year blocks, many candidate comparisons failed because at least one conditional fit contained too few exceedances.
- Twenty-year blocks were reliable for winter but produced substantial failure rates for several other seasons.
- All candidate comparisons fitted successfully with 25-year blocks.
- Increasing the length to 30 years also produced stable fits, but left only two to five candidate changepoints per season.
- We therefore use 25 season-years on either side of each candidate throughout the primary analysis.
- This was the shortest common block length that produced stable fits across all seasons while retaining a useful range of candidate changepoints. **Present the alternative block lengths as a sensitivity analysis or in the supplementary material(?).**

4.3 | Screening

- We first perform a computationally less intensive screening step, in which the observed discrepancy between the two estimated dissimilarity matrices is calculated at every feasible candidate season-year without generating a permutation distribution.
- Figure 2 presents the screening statistics for each season using the Frobenius, infinity and spectral norms.
- As discussed in Section 2, these norms provide complementary summaries of the difference between the dissimilarity matrices on either side of a candidate change.
- The three norms have different numerical scales, and their absolute values should therefore not be compared directly.
- Instead, we compare the timing and relative prominence of features in their respective screening curves.
- Red points in Figure 2 identify local peaks.
- For a given norm (T), an interior candidate season-year (t) is classified as a local peak when

$$T(t) > T(t-1) \quad \text{and} \quad T(t) \geq T(t+1).$$

The first and final candidate years are not classified as local peaks because they cannot be compared with candidates on both sides.

- For winter, all three norms increase during the late 1990s and exhibit a prominent shared peak in 1998.
- A smaller common peak is present around 1992.
- The agreement among the norms makes the late-1990s period a particularly notable candidate region.
- For spring, the clearest screening peak occurs around 1992, for all norms.
- Smaller local peaks occur around 1989 and 1997.
- For summer, the screening curves contain several prominent peaks.
- The largest norm values occur in 1988, with further substantial peaks in 1992 and 1997.
- For Autumn, the screening results are less concentrated around a single candidate year.
- The Frobenius curve has local peaks around 1990 and 1996, while the infinity and spectral curves identify broadly similar candidate regions during the late 1980s or early 1990s and the mid-1990s.
- The precise locations of the peaks are less consistent across norms than for winter, spring or summer.
- Across the four seasons, the largest Frobenius norm screening value occurs for summer around 1988. **Cite exact value**
- The clearest agreement among all three norms occurs around 1992 for spring and around 1998 for winter.
- Although the screening step highlights several candidate regions of particular interest, the total number of feasible candidate season-years is sufficiently small to apply the permutation test at every candidate.
- We therefore retain the complete candidate range rather than restricting the formal analysis to the local screening peaks.

4.4 | Change point detection

- We next apply the permutation test at every feasible candidate season-year.
- For each candidate and norm, the null distribution is estimated using 1000 permutations of complete season-years between the two 25-year blocks.
- Figure 3 shows the resulting pointwise Monte Carlo p-values for the Frobenius, infinity and spectral norms.
- A candidate season-year (t) represents a possible change between season-years (t) and ($t + 1$), and the horizontal dashed line denotes the pointwise 5% significance level.
- The p-values in Figure 3 are pointwise: each tests a change at one specified candidate season-year.
- For spring, none of the three norms produces a pointwise p-value below 0.05.
- The smallest values occur around 1992, in agreement with the dominant feature identified during screening, but remain above the pointwise significance threshold.
- Also for Autumn, none of the three norms produces a pointwise p-value below 0.05, although the infinity norm came very close for 1986, the year with the highest test statistic in our screening step.
- For winter, the infinity norm provides pointwise evidence of a change over the period 1996–1999.
- The spectral norm supports a narrower region around 1997–1998, while the Frobenius norm is significant in 1998.
- Thus, all three norms identify 1998, and the overall pattern provides consistent evidence for a change in the winter extremal dependence structure during the late 1990s.
- For summer, the infinity norm produces pointwise-significant p-values over an extended period from approximately 1987 to 1993.

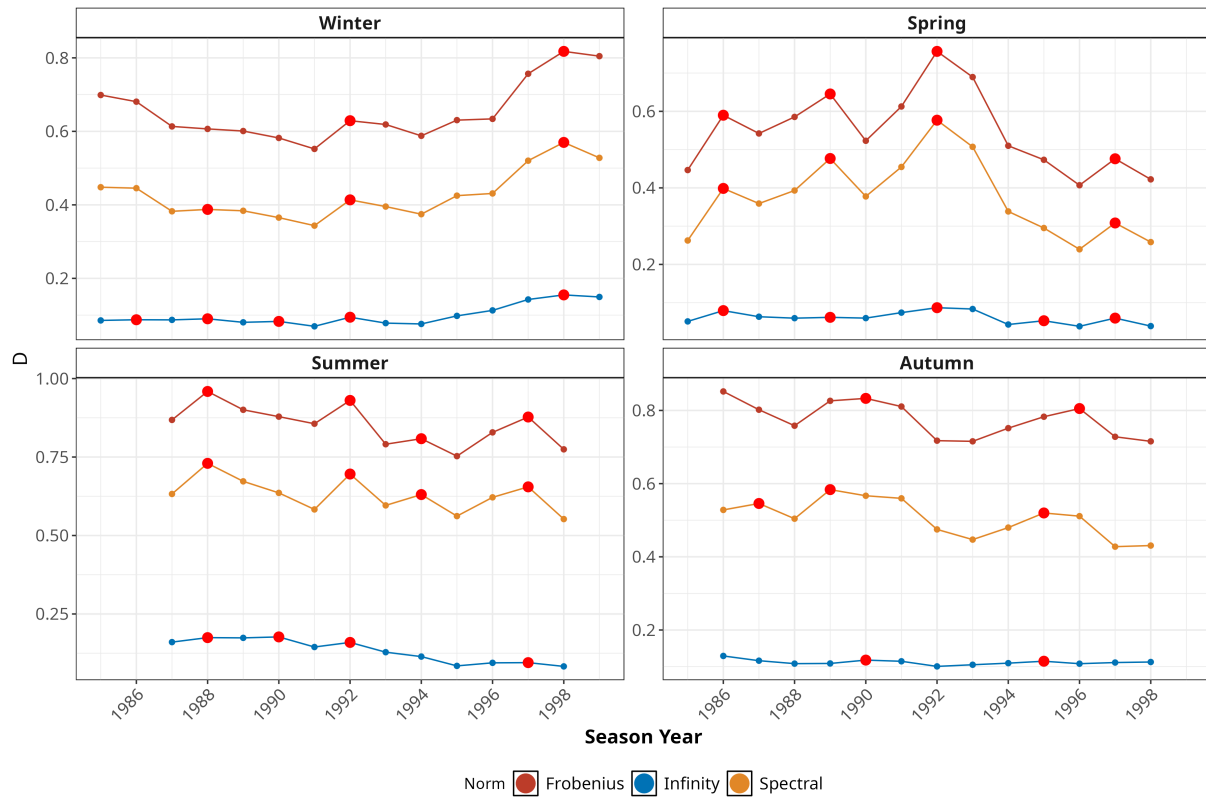


FIGURE 2 Screening discrepancies between the estimated extremal-dependence dissimilarity matrices on either side of each candidate season-year, calculated using 25 season-years per temporal block. Results are shown for the test statistic D using the Frobenius, infinity and spectral norms. A candidate (t) represents a possible change between season-years (t) and ($t + 1$). Red points denote interior local peaks satisfying $(T(t) > T(t - 1))$ and $(T(t) \geq T(t + 1))$.

- The Frobenius and spectral norms support narrower subsets of this region, with particularly small p-values around 1988-1990 and 1992.
- The agreement among the norms indicates that the early-1990s signal is not confined to a single choice of discrepancy statistic.
- After approximately 1994, the summer p-values generally increase, particularly for the infinity norm.
- These results provide no pointwise evidence for a change at these later candidates.
- However, the relatively large p-values should not be interpreted as affirmative evidence that the dependence structure is unchanged; they indicate only that the observed discrepancies are not unusual under the corresponding permutation null distributions.
- The infinity norm identifies broader pointwise-significant candidate ranges than the Frobenius and spectral norms for both summer and winter.
- As discussed in Section ??, this may indicate that some changes are more pronounced in particular entries of the dissimilarity matrix than across its entire structure.
- Nevertheless, the principal summer and winter candidate regions are supported by more than one norm.
- In particular, all three norms provide pointwise evidence around 1990 for summer and around 1998 for winter.
- Taken together, the results suggest candidate changes in the summer extremal dependence structure during the late 1980s or early 1990s and in the winter structure during the late 1990s.

- No comparable evidence is found for spring or autumn.

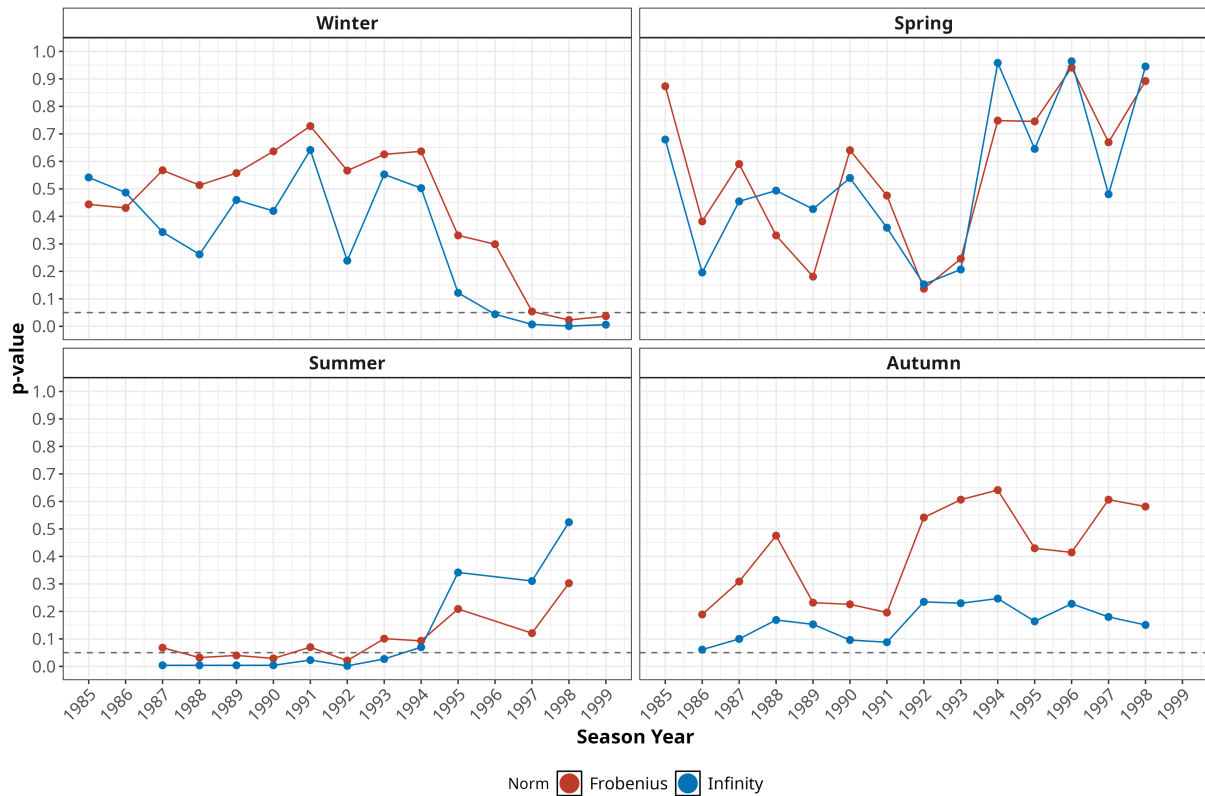


FIGURE 3 Pointwise permutation-test p-values for candidate changes in the seasonal extremal dependence structure, calculated using 25 season-years per temporal block and 200 permutations at each candidate. Results are shown for the Frobenius, infinity and spectral norms. A candidate (t) represents a possible change between season-years (t) and ($t + 1$). The horizontal dashed line denotes the pointwise 5% significance level.

4.5 | Clustering Results

- We finally examine how the detected changes affect the spatial clustering of the fitted extremal dependence structure.
- For each fitted conditional extremes model, the aggregated dissimilarity matrix is used to partition the 40 stations into three clusters using the clustering procedure described in Section ??.
- As the changepoint analysis provides no consistent evidence of a change in spring or autumn, we refit the conditional extremes model to the complete time series for these seasons.
- For summer, the pointwise tests identify a broader candidate change region during the late 1980s and early 1990s.
- We use 1990 as a representative division within this region and fit separate models to season-years 1960–1990 and 1991–2020.
- For winter, the three norms provide their clearest common evidence at season-year 1998.
- We therefore fit separate models to winter season-years 1960–1998 and 1999–2020.

- These divisions treat a candidate (t) as a change between season-years (t) and ($t + 1$). **Confirm or update the selected divisions after completing the multiplicity-adjusted changepoint analysis.**
- Figure 4 presents the resulting spatial clusterings.
- Although station coordinates and elevation do not directly determine the clustering assignments, the fitted clusters exhibit substantial spatial coherence.
- Neighbouring stations are frequently assigned to the same cluster, suggesting that the estimated extremal dependence structure contains a strong geographical component.
- The spring clustering broadly separates groups of stations along the northern and Mediterranean coasts (in blue) from groups in the western (orange) and central interior (red).
- Nevertheless, the boundaries are not determined solely by physical proximity, with some geographically separated stations assigned to the same dependence cluster.
- The autumn clustering exhibits a stronger broadly coastal–interior structure.
- Several stations along the northern and eastern coasts are grouped together, similar to the spring clustering, while the other two clusters comprise locations in the west/south-west and elsewhere, respectively.
- For summer, the two fitted periods retain some common large-scale structure.
- However, a number of stations change their clustering relationships after 1990, most visibly in northwestern Spain, along the southern coast and across parts of the western interior.
- The summer changepoint therefore corresponds to a reorganization of particular regional relationships rather than a complete replacement of the earlier clustering structure.
- The winter clustering, in contrast, changes more extensively across the 1998 division.
- In particular, the post-1998 fit produces a different grouping of many stations along the northern coast, in the western and central interior, and along the Mediterranean coast.
- This broader spatial reorganisation, as opposed to the more localised changes seen in summer, is consistent with the late-1990s signal being identified by all three norms in the changepoint analysis.

5 | DISCUSSION

ACKNOWLEDGMENTS

This is acknowledgment text [Elbaum, Malishevsky, and Rothermel \(February 2002\)](#). Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here.

Author contributions

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Financial disclosure

None reported.

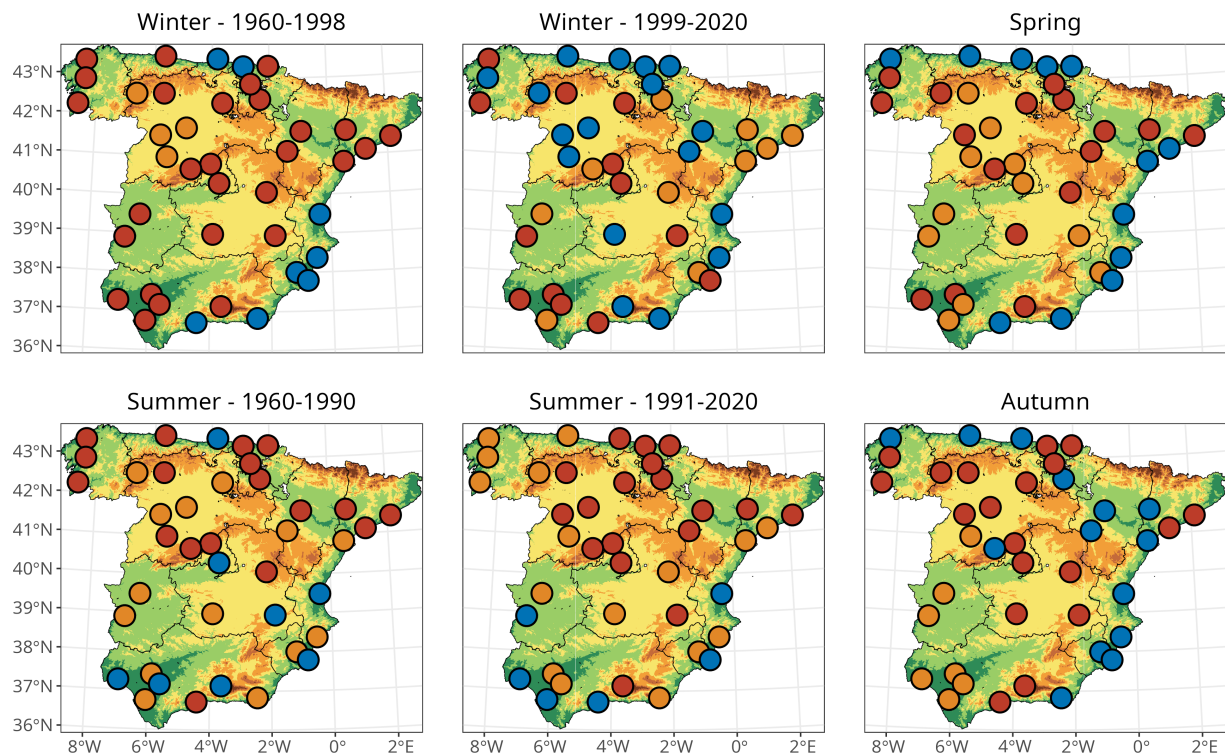


FIGURE 4 Spatial clustering of the 40 Spanish stations obtained from the fitted conditional extremes dissimilarity matrices. Spring and autumn are fitted using their complete time series. Summer is fitted separately for season-years 1960–1990 and 1991–2020, while winter is fitted separately for season-years 1960–1998 and 1999–2020. Three clusters are used in each fit. Colours distinguish cluster membership within each panel. Background shading represents elevation.

Conflict of interest

The authors declare no potential conflict of interests.

SUPPORTING INFORMATION

The following supporting information is available as part of the online article:

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APPENDIX

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References

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